MIT EECS / CSAIL-style Doctoral Research Portfolio

Realistic S3 portfolio for PhD-level AI systems research readiness

Prepared for	MIT EECS / CSAIL-style doctoral research review
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Positioning	PhD-style research proposal and evidence package, not a claim of affiliation
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This doctoral portfolio is built to look like a research trajectory, not a graphic showcase. It avoids fake publications, fake supervisor approval, and overclaiming. The strongest doctoral presentation is a clear problem gap, early evidence, a testable method, a publication roadmap, and honest risks.

Doctoral positioning

A PhD portfolio should not simply list projects. It should show that the applicant can turn several projects into one coherent research agenda. The five projects are reframed as preliminary evidence for a doctoral theme in responsible AI systems for education analytics, student feedback intelligence, and socio-technical decision support.

Proposed dissertation direction

Working title: Responsible AI Systems for Student Support: Interpretable Risk Prediction, Feedback Intelligence, and Socio-Technical Evaluation. The dissertation would study how AI systems can support education institutions without creating opaque, unfair, or overconfident decision pipelines.

Research problem gap

Many AI-for-education prototypes focus on predictive performance but under-document dataset assumptions, uncertainty, fairness, human review, and institutional consequences. A stronger doctoral portfolio should ask how risk prediction, NLP feedback analysis, and dashboard systems can be evaluated as socio-technical systems, not just as isolated machine learning models.

Research questions

RQ1	How can early academic risk models be designed to communicate uncertainty, limitations, and human-review requirements?
RQ2	How can NLP feedback systems identify actionable themes while minimizing bias and misinterpretation of student voice?
RQ3	How should dashboards and evidence hubs present AI outputs so that reviewers understand methods, risks, and reproducibility?
RQ4	What evaluation framework can connect accuracy, fairness, interpretability, and real-world support outcomes?

Preliminary evidence matrix

Evidence item	Realistic doctoral-level use	What must be shown
AI Student Performance Prediction	Preliminary evidence for a doctoral theme on responsible AI for educational risk prediction.	Problem gap, causal limits, responsible AI framing, longitudinal evaluation plan, privacy/ethics.
Malaysia Housing and Cost of Living Analysis	Socio-technical evidence for AI-assisted student support and equity-aware decision systems.	Research question, stakeholder map, bias risks, generalization constraints, decision-support evaluation.
Text Sentiment Analysis for Student Feedback	Prototype evidence for human-centered NLP and institutional feedback intelligence.	Annotation protocol, disagreement handling, interpretability, harm analysis, domain adaptation plan.
Computer Vision Image Classification	Supporting module for smart campus safety, not the dissertation core unless expanded.	Data provenance, failure modes, safety risks, benchmark plan, human-in-the-loop safeguards.
AI Portfolio Website	Research artifact registry and reproducibility control center.	Versioned evidence, transparent limitations, literature map, publication roadmap, code/data access plan.

Methodology architecture

Layer	Doctoral method
Data layer	Document provenance, consent assumptions, anonymization, dataset shift, and missing-data risks.
Model layer	Compare baselines, calibrated models, interpretable models, and error clusters rather than claiming one perfect model.
Human layer	Define how advisors, reviewers, or student-support staff interpret outputs and when they should override the model.
Evaluation layer	Combine accuracy, calibration, fairness checks, qualitative error analysis, and reproducibility evidence.
Governance layer	Use model cards, data sheets, ethics notes, access controls, and change logs.

Publication roadmap

Paper idea	Target contribution	Evidence needed
Paper 1: Interpretable academic risk prediction	A calibrated and explainable pipeline for early support, with honest limitations.	Reproducible code, baseline comparison, fairness note, case-based error analysis.
Paper 2: Human-centered NLP for student feedback	A feedback intelligence workflow that balances topic discovery with student voice protection.	Annotation notes, topic model, sentiment baseline, qualitative review, ethics statement.
Paper 3: Socio-technical dashboard evaluation	A dashboard/evidence hub framework for communicating AI outputs responsibly.	Prototype, user tasks, evidence traceability, reproducibility checklist, failure mode review.

Advisor or lab fit section template

Use fit areas, not invented promises. A realistic MIT-style section should mention research alignment with AI systems, human-centered AI, machine learning, NLP, computer vision, HCI, data systems, or societal impact. Do not write that a professor has accepted the student unless there is written proof. Use "potential alignment" or "research fit to explore".

Human realism checklist

Risk	Realistic fix
Looks AI-generated	Add specific datasets, dates, repo links, limitations, and manual analysis notes.
Looks inflated	Remove fake publications, fake affiliation, fake supervisor approval, and unsupported metrics.
Looks like a design portfolio only	Add research questions, methodology, ethics, reproducibility, and publication roadmap.
Looks too generic	Tie all projects to one coherent dissertation theme and evidence chain.

Sources and alignment notes

MIT EECS official admissions material states that graduate application is for the doctoral program only and that PhD students earn a master degree during the PhD path. MIT EECS academic progress materials describe doctoral research milestones such as research qualifying evaluation and thesis proposal. This portfolio uses those facts to structure the package realistically. It does not claim admission, endorsement, or affiliation.